

Unit 1, Lesson 6: Writing and Solving One-Step Equations in Real-World Contexts

Benchmarks

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Target

- ☐ I can write and solve one-step addition, subtraction, multiplication, and division equations that represent a real-world context.

1.6.1 Warm-Up: Would You Rather?

The number of rooms and rate per night at two different Florida hotels are shown.

Downtown Miami Hotel	Luxury Key West Resort
 527 rooms at \$139 per night	 40 cottages at \$1,049 per night

1. From which business would you rather earn the revenue?
2. Explain your reasoning using mathematics.

1.6.2 Collaboration: Writing and Solving Equations in Real-World Context

Work with your partner to complete the following.

1. Diandra’s cat weighs 6 pounds fewer than Kim’s puppy. Kim’s puppy weighs 17 pounds.
- a. Select all of the equations that could be used to find the weight of Diandra’s cat.
- ☐ $x + 6 = 17$

☐ $x - 6 = 17$

☐ $x = 17 + 6$

☐ $x = 17 - 6$

☐ $x - 17 = 6$

☐ $x + 17 = 6$
- b. What does the variable, x , represent in this situation?
- c. What is the weight of Diandra’s cat?

2. Four equations are shown.

$6x = 42$

$42x = 6$

$\frac{x}{42} = 6$

$\frac{x}{6} = 42$

- a. Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity.

Situation 1	Situation 2
At a restaurant, 6 friends decided to split the bill equally. If each person paid \$42, what was the total bill?	A small theater has a total of 42 seats. If the theater has 6 rows of seats with the same number of seats in each row, how many seats are in each row?

- b. Write the matching equation in the space provided.

Situation 1	Situation 2
Equation:	Equation:

- c. What does x represent in each equation?

Situation 1	Situation 2
In this situation, the quantity x represents ...	In this situation, the quantity x represents ...

- b. Solve either of the equations for d . Use the other equation to verify your solution is correct.

1.6.4 Collaboration: Writing Real-World Scenarios to Represent Algebraic Equations

1. For each equation, create a real-world problem that the algebraic equation could represent.

Equation	Your Real-World Scenario	What does the variable represent in your scenario?
$r - 10 = 12$		
$21y = 105$		
$\frac{a}{10} = 3$		
$17 + s = 26$		

2. Pick two equations from the table and compare your scenarios and solutions with your partner. Use them to complete the table below.

Equation	Your Partner's Scenario	Does the equation represent your partner's scenario?	Similarities between scenarios for the equation
		☐ Yes ☐ No	
		☐ Yes ☐ No	

1.6.5 Wrap-Up: Ticket Out the Door

1. Four equations are shown.

$3x = -15$

$x - 3 = 15$

$\frac{x}{3} = 15$

$x + 3 = -15$

Match an equation to each situation it represents. Then, complete the table. Not all equations will be used.

Situation	Equation	What does x represent?	Solution
a. On Tuesday, the temperature in Nunavut rose 3° from the previous day to -15° . What was the temperature on Monday?			
b. Tyanna is re-vamping her wardrobe. She decides to donate 3 old t-shirts. Now, she has 15 t-shirts. How many t-shirts did she have originally?			
c. A shipment of oranges was split into 3 equally sized crates. Each crate weighs 15 pounds. How much did the original shipment weigh?			